

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202506

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost,	Accuracy=0.9893713857357036,	Precision=0.22254169062679702,	Recall=0.911660777385159,
F1=0.35775363993529,	ROC_AUC=0.9860349941936518		
algorithm=stacked_ensemble,	Accuracy=0.9547509408561026,	Precision=0.06386020651310564,	Recall=0.9469964664310954,
F1=0.11965175980355681,	ROC_AUC=0.9822966410407276		
algorithm=easy_ensemble,	Accuracy=0.7546889820395924,	Precision=0.010002632271650434,	Recall=0.7608951707891637,
F1=0.01974569018217386,	ROC_AUC=0.8490409780236461		
algorithm=gradient_boosting,	Accuracy=0.9843764342318636,	Precision=0.16748869708179204,	Recall=0.9599528857479388,
F1=0.2852143482064742,	ROC_AUC=0.9921433167948055		
algorithm=random_forest,	Accuracy=0.998978826913074,	Precision=0.8523002421307506,	Recall=0.8292108362779741,
F1=0.8405970149253731,	ROC_AUC=0.9839299458083961		